



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to DEVA SERVICE

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

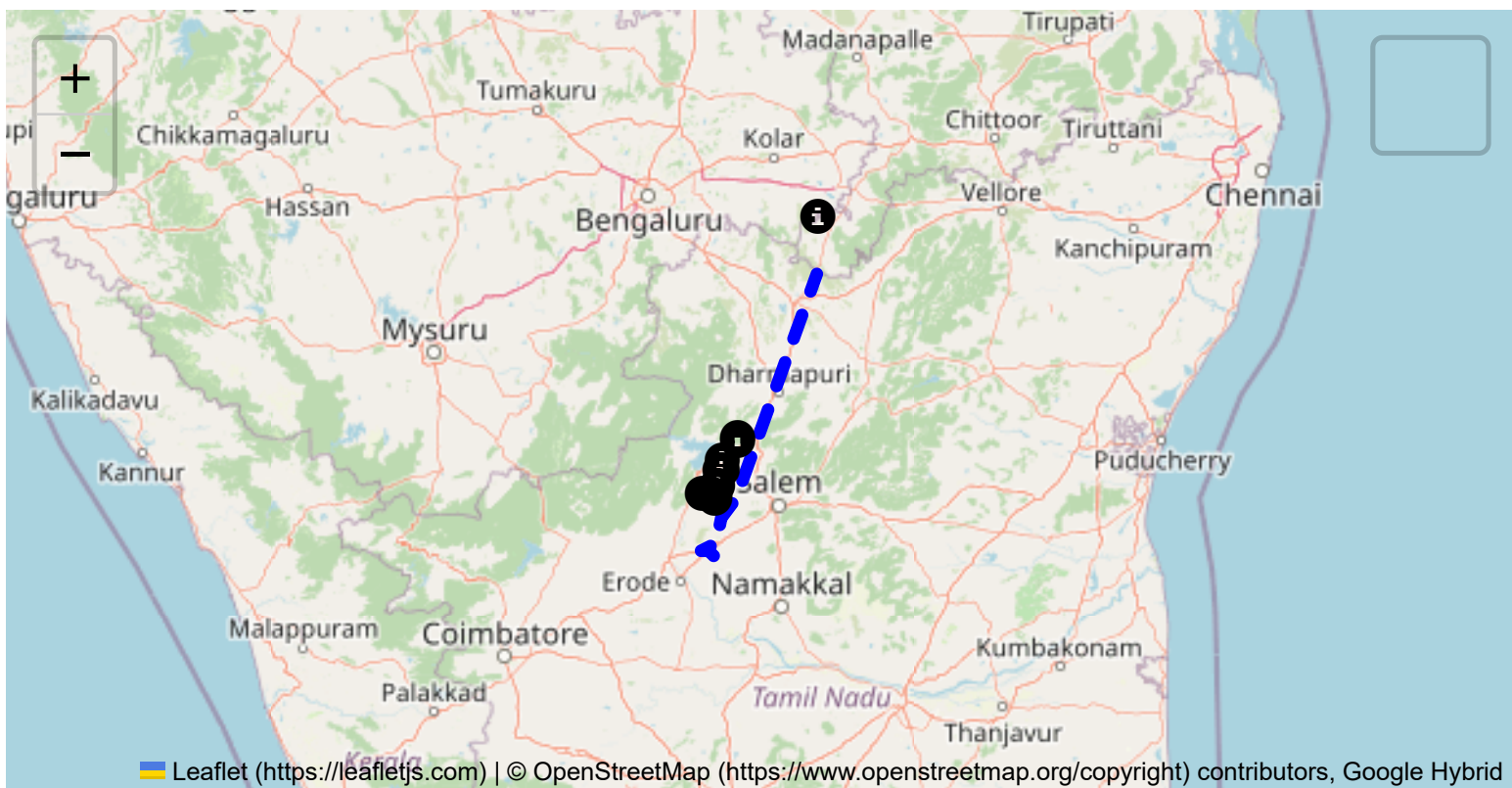
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

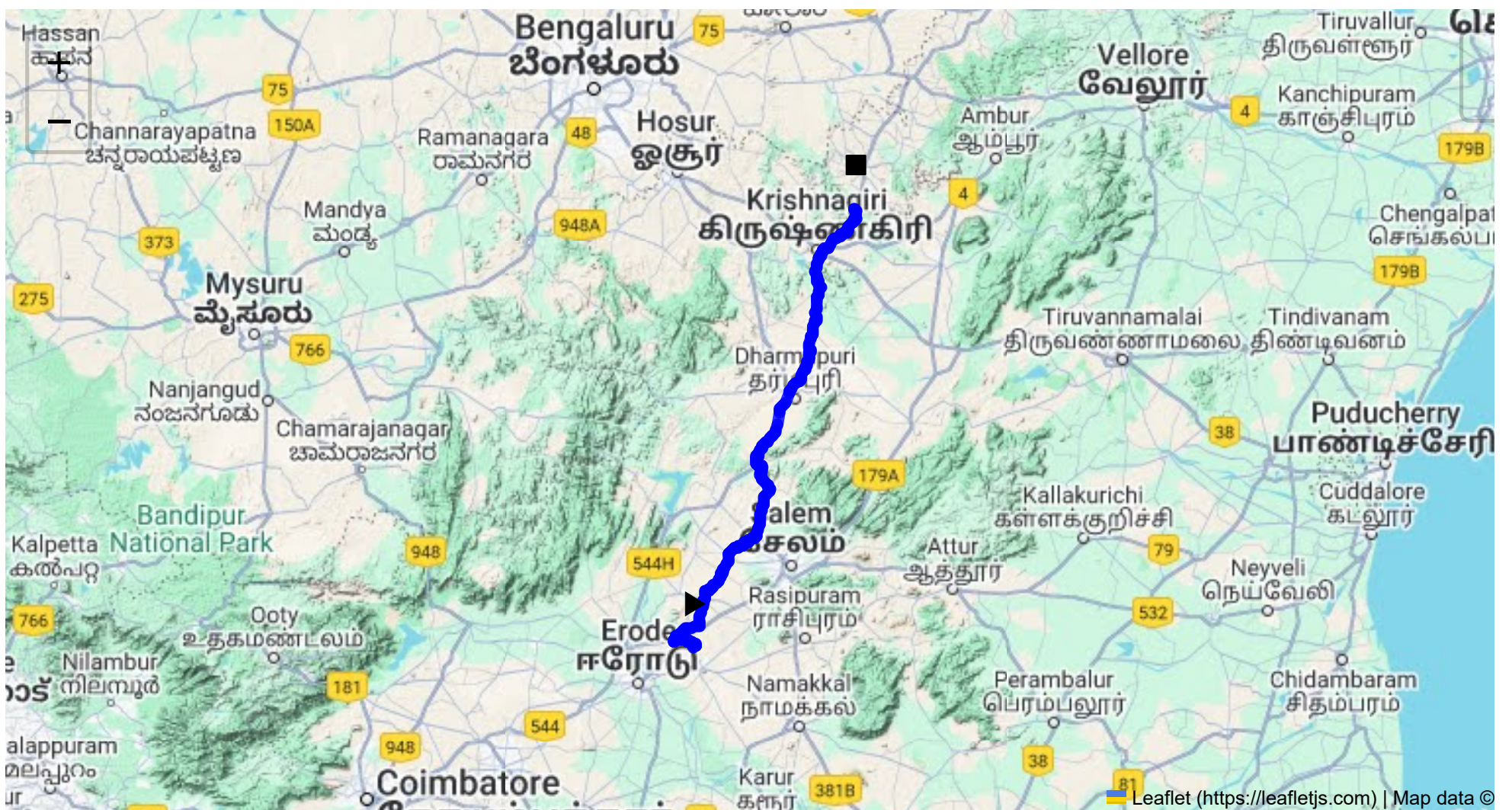
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 170.39 km
Estimated Duration: 3.3 hours
Adjusted Duration (Heavy Vehicle): 4.1 hours
Start: (11.4381, 77.8734)
End: (12.642548, 78.322032)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Guruvinnayanapalli traverses approximately 170.39 kilometers through several districts in Tamil Nadu, following local roads and state highways, particularly SH 86. It is a mixed route combining urban, semi-urban, and rural environments. The key locations along the route include Sangagiri, Bhavani, Erumaipatti, and Tharamangalam, providing vital connectivity across these regions.

Weather Conditions and Hazards

Typical weather in this region includes hot summers, with temperatures potentially reaching up to 40°C (104°F). The monsoon season, from June to September, can bring heavy rains causing waterlogging in lower areas and potentially flooding some road sections. The driver should be cautious of reduced visibility and slippery roads during this period.

Traffic Patterns

- **Peak Hours:** Morning (7:30-9:30 AM) and evening (5:00-7:00 PM) typically exhibit heightened traffic congestion, especially around major towns such as Sankari and Tharamangalam.
- **Congestion Areas:** Town centers and intersections, particularly in urban settlements like Sangagiri and Erumaipatti, can become congested. Roads near markets and schools may also see temporary jams.

Road Quality and Infrastructure

Overall, the road quality varies:

- **Urban Sections:** Generally well-paved, but high vehicular movements can lead to wear and minor potholes.
- **Rural Sections:** Some stretches may be narrower with less maintenance, potentially affecting a smooth drive.
- **Infrastructure:** Signage is generally adequate, but intersections may lack sophisticated traffic management systems.

Alternative Routes for Emergencies

- **Route Diversions:** For bypassing congested towns or closed roads due to incidents, explore connecting state highways like SH 20 and SH 79, which intersect the planned route at various points.
- **Local Roads:** Know side roads linking major highways generally used by locals for bypassing traffic, particularly in areas like Sankari.

Local Regulations on Hazardous Transport

- Hazardous material transport in Tamil Nadu requires compliance with specific routes and timings to minimize public and environmental risk.
- Nighttime rest stops for hazardous materials may be regulated to designated areas.

Historical Incidents

There have been occasional incidents involving heavy vehicles, primarily due to brake failures on inclines or congestion-based collisions in more populated areas like Sankari. No major hazardous material spills reported recently, but caution is advised.

Environmental Considerations

- **Sensitive Areas:** The route goes through regions that may have agricultural fields and local water resources.
- **Environmental Risks:** Spillage or accidental discharge should be carefully managed to prevent harm to these surroundings.

Communication Coverage

- Good mobile network coverage in towns such as Sangagiri, Erumaipatti, and Tharamangalam.
- Possible dead zones might exist in rural or less developed segments; carrying a GPS device independent of mobile data is recommended.

Emergency Response Times

- **Urban Areas:** Approximately 15-30 minutes due to proximity to local emergency services.
- **Rural Areas:** Could range from 30 minutes to over an hour depending on location and road conditions.

Overall Summary of Risk Assessment

While the route is generally navigable for heavy vehicles carrying hazardous materials, notable risks include weather-induced hazards, regional traffic congestion, and sections with potentially lower road quality. Proper adherence to regulatory guidance for hazardous materials, maintaining heightened vigilance during monsoon periods, and readiness for alternative routing will mitigate the bulk of identified risks. Emergency response infrastructure is reliable in urban settings but less so in rural areas, making preventive measures and preparedness essential for ensuring route safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45764, 77.85219	30 KM/Hr
5	Turn	High	11.46303, 77.84945	15 KM/Hr
6	Turn	High	11.45571, 77.81702	15 KM/Hr
7	Turn	Medium	11.45654, 77.81706	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
8	Turn	Medium	11.46916, 77.82431	30 KM/Hr
9	Blind Spot	Blind Spot	11.49395, 77.88537	10 KM/Hr
10	Turn	High	11.49465, 77.88488	15 KM/Hr
11	Turn	Medium	11.56096, 77.89931	30 KM/Hr
12	Turn	Medium	11.57649, 77.90610	30 KM/Hr
13	Turn	Medium	11.59214, 77.90656	30 KM/Hr
14	Turn	Medium	11.63833, 77.94818	30 KM/Hr
15	Turn	High	11.64243, 77.95198	15 KM/Hr
16	Turn	Medium	11.64721, 77.95118	30 KM/Hr
17	Turn	Medium	11.69559, 77.97112	30 KM/Hr
18	Turn	Medium	11.71386, 77.99552	30 KM/Hr
19	Turn	Medium	11.71735, 78.00019	30 KM/Hr
20	Turn	Medium	11.71915, 78.01106	30 KM/Hr
21	Turn	Medium	11.72492, 78.01660	30 KM/Hr
22	Turn	Medium	11.72765, 78.02291	30 KM/Hr
23	Turn	Medium	11.73764, 78.03441	30 KM/Hr
24	Turn	Medium	11.73752, 78.03679	30 KM/Hr
25	Turn	High	11.73652, 78.03866	15 KM/Hr
26	Turn	Medium	11.74110, 78.04575	30 KM/Hr
27	Turn	Medium	11.74346, 78.04719	30 KM/Hr
28	Turn	Medium	11.78094, 78.05792	30 KM/Hr
29	Turn	Medium	11.86805, 78.08353	30 KM/Hr
30	Turn	Medium	11.87022, 78.08380	30 KM/Hr
31	Turn	Medium	11.91368, 78.05872	30 KM/Hr
32	Turn	Medium	11.92566, 78.06503	30 KM/Hr
33	Turn	Medium	11.93097, 78.06492	30 KM/Hr
34	Turn	Medium	11.93429, 78.06326	30 KM/Hr
35	Turn	Medium	11.93699, 78.05412	30 KM/Hr
36	Turn	Medium	11.95551, 78.04706	30 KM/Hr
37	Turn	High	11.95810, 78.04971	15 KM/Hr
38	Turn	High	11.96384, 78.05141	15 KM/Hr
39	Turn	Medium	11.96491, 78.05001	30 KM/Hr
40	Turn	High	11.96657, 78.04930	15 KM/Hr
41	Turn	Medium	11.96891, 78.05109	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
42	Turn	Medium	12.22440, 78.19131	30 KM/Hr
43	Turn	Medium	12.24696, 78.19569	30 KM/Hr
44	Turn	Medium	12.25029, 78.19959	30 KM/Hr
45	Turn	Medium	12.25296, 78.20042	30 KM/Hr
46	Turn	Medium	12.29079, 78.19897	30 KM/Hr
47	Turn	Medium	12.32029, 78.20737	30 KM/Hr
48	Turn	Medium	12.40130, 78.21824	30 KM/Hr
49	Turn	Medium	12.44346, 78.21837	30 KM/Hr
50	Turn	High	12.50722, 78.22222	15 KM/Hr
51	Turn	Medium	12.51441, 78.23022	30 KM/Hr
52	Turn	Medium	12.53163, 78.23950	30 KM/Hr
53	Turn	Medium	12.53485, 78.24863	30 KM/Hr
54	Turn	Medium	12.55970, 78.26809	30 KM/Hr
55	Turn	Medium	12.56883, 78.28466	30 KM/Hr
56	Turn	Medium	12.56985, 78.29059	30 KM/Hr
57	Blind Spot	Blind Spot	12.57704, 78.30192	10 KM/Hr
58	Turn	Medium	12.57745, 78.30156	30 KM/Hr
59	Turn	High	12.58017, 78.30126	15 KM/Hr
60	Turn	Medium	12.58139, 78.30311	30 KM/Hr
61	Turn	Medium	12.60199, 78.31816	30 KM/Hr
62	Turn	Medium	12.60449, 78.31855	30 KM/Hr
63	Turn	Medium	12.61239, 78.32541	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium
3	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
4	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium
5	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
6	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
7	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
8	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
9	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
10	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
11	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
12	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
14	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
15	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
16	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
17	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
18	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
19	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
20	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
21	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
22	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
23	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
24	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
25	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
27	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
28	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
29	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
30	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
31	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
36	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
13	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
26	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
32	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
33	school	Nalanda Metric Higher Secondary School	12.488407, 78.21856	30 km/h	Medium
34	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium
35	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium
37	college	Government Polytechnic College	12.5352064, 78.2460463	30 km/h	Medium
38	college	Government Boys Arts and Science college	12.5323457, 78.2457274	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45764, 77.85219



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45571, 77.81702



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45654, 77.81706



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46916, 77.82431



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49395, 77.88537



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49465, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56096, 77.89931



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57649, 77.90610



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59214, 77.90656



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63833, 77.94818



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64243, 77.95198



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64721, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69559, 77.97112



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71386, 77.99552



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71735, 78.00019



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71915, 78.01106



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72492, 78.01660



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72765, 78.02291



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73764, 78.03441



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73752, 78.03679



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73652, 78.03866



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74110, 78.04575



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74346, 78.04719



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.78094, 78.05792



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86805, 78.08353



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87022, 78.08380



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91368, 78.05872



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.92566, 78.06503



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93097, 78.06492



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93429, 78.06326



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93699, 78.05412



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95551, 78.04706

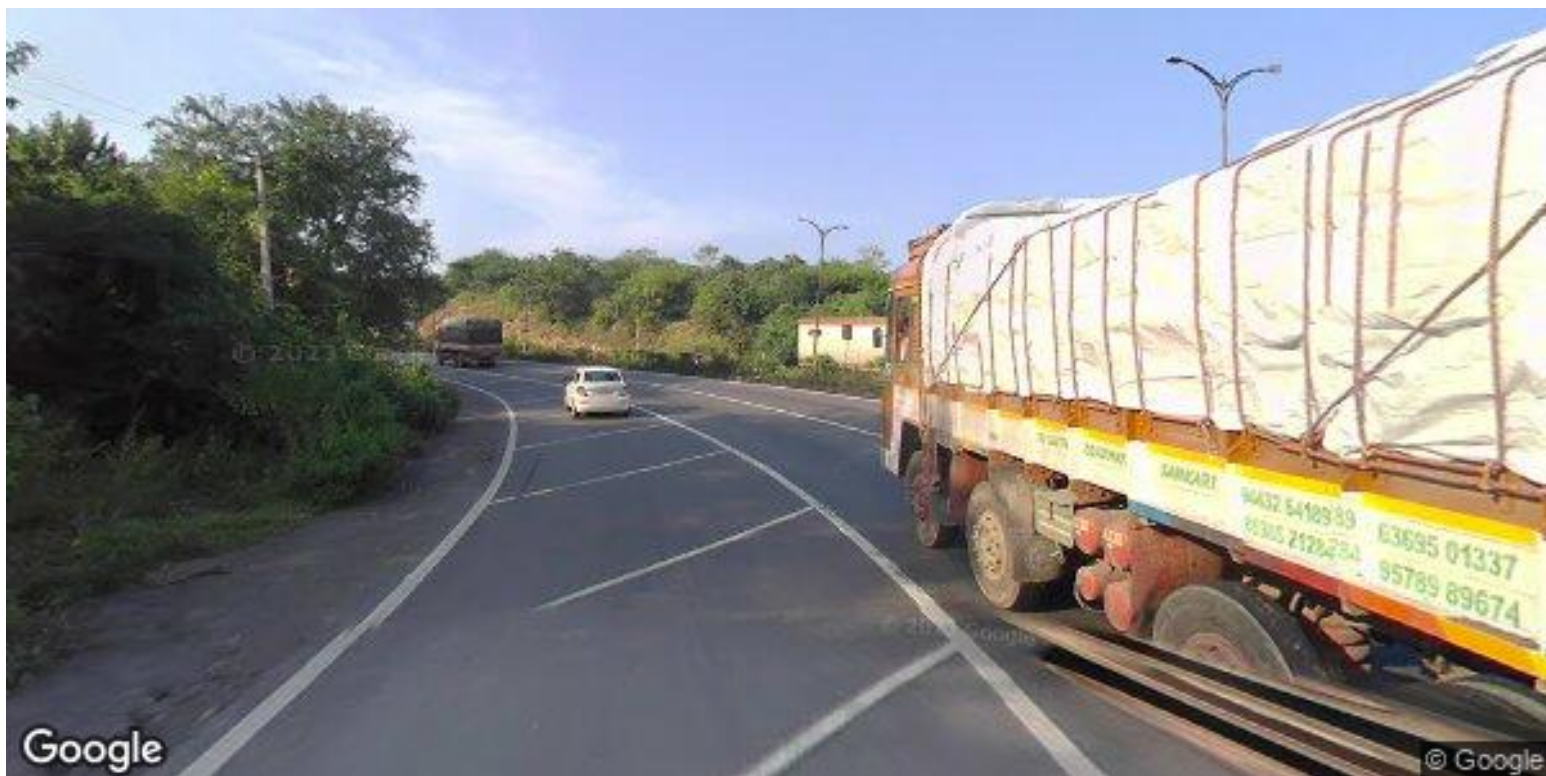


Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.95810, 78.04971



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96384, 78.05141



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96491, 78.05001



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96657, 78.04930



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96891, 78.05109



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.22440, 78.19131



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.24696, 78.19569



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25029, 78.19959



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25296, 78.20042



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.29079, 78.19897



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.32029, 78.20737



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.40130, 78.21824



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.44346, 78.21837



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.50722, 78.22222



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.51441, 78.23022



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.53163, 78.23950



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

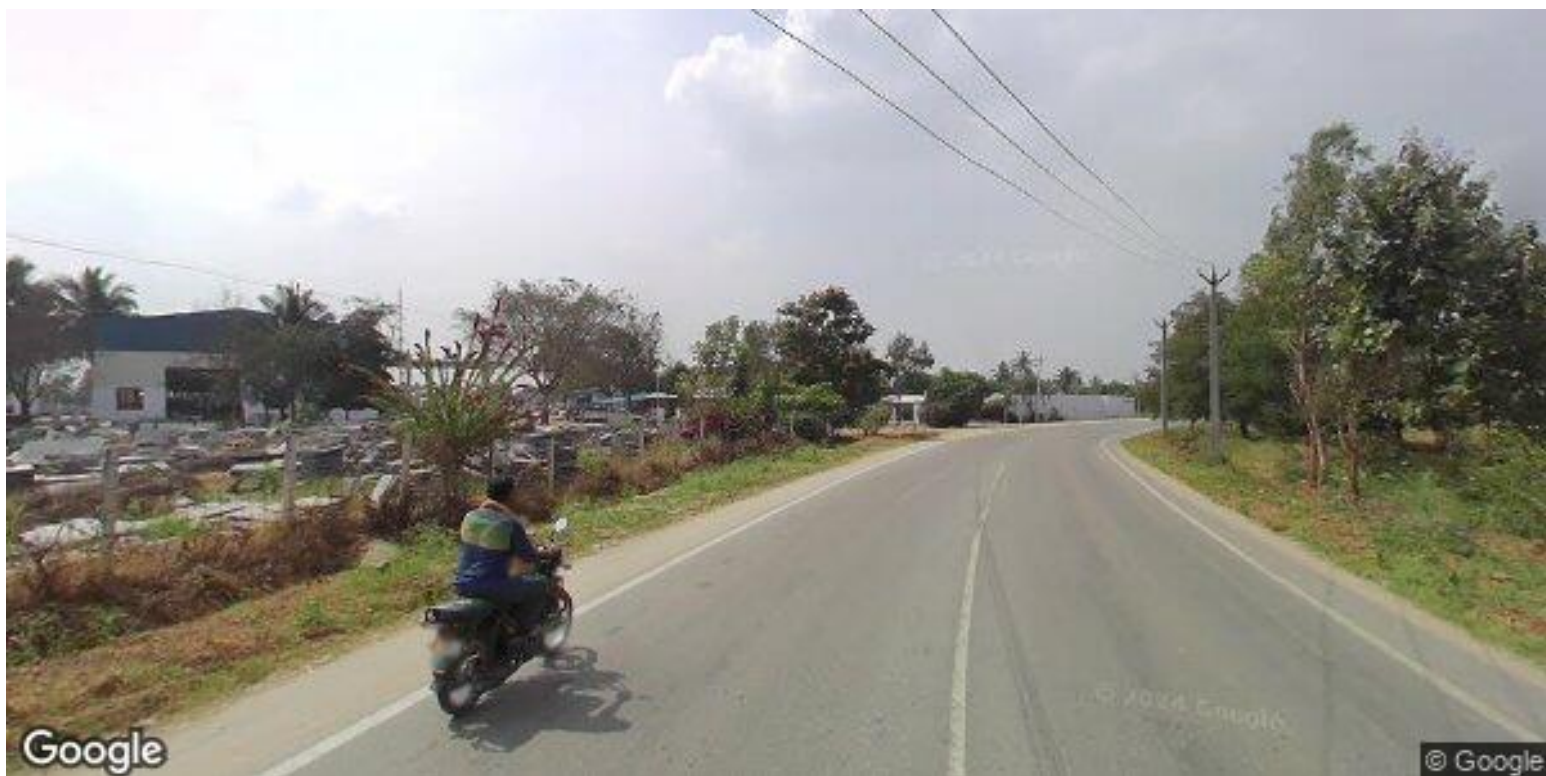
Coordinates: 12.53485, 78.24863



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.55970, 78.26809



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.56883, 78.28466



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.56985, 78.29059



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.57704, 78.30192



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.57745, 78.30156



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.58017, 78.30126



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.58139, 78.30311



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.60199, 78.31816



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.60449, 78.31855



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.61239, 78.32541
